

Cross-Asset Portfolio Construction as Cashflow Engineering

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Abstract

This paper studies a practical version of cross-asset allocation: an investor starts with SPY exposure and asks whether stocks inside the index, gold, copper, VIX futures, Treasuries, and SPY options improve the portfolio after funding, transaction costs, option premia, drawdowns, and tail losses are charged. The answer is conditional. A portfolio is not just a weight vector. It is a financed domestic-cashflow process whose instruments differ by payoff map, margin convention, hedgeability, and friction.

The paper develops the accounting and objective before showing results. Each asset is converted into domestic returns and exposures. Pricing measures are separated from physical expected returns. Sortino ratio, lower partial moments, maximum drawdown, CVaR, and stress-window survival are the main tests; Sharpe is reported as a diagnostic. The empirical experiment builds a daily execution panel from local repository data and then aggregates executed P&L back to the monthly evaluation horizon. The outer allocator uses a monthly panel of SPY, an equal-weight large-cap constituent basket, gold and copper futures proxies, a VIX-futures proxy, TLT, 95/105 target SPY option overlays from OPRA month-end slices, and diagnostic long-dated option proxies kept outside the headline optimizer. The inner layer applies the monthly targets on daily close/settlement returns with explicit transaction-cost, option-lifecycle, delta-hedge, futures-roll, and drawdown-control ledgers. A rolling out-of-sample test from February 2018 through May 2026 is then checked with combinatorial purged cross-validation, probability-of-backtest-overfitting diagnostics, daily block bootstrap, EWMA volatility-clustered Monte Carlo, cost multipliers, and a conservatively gated ML forecast arm.

The experiment supports the theory but also disciplines it. Multi-asset diversification and a downside objective improve the tested path: the no-option downside-score portfolio reaches a 2.00 Sortino ratio and a -19.87% maximum drawdown versus SPY's 1.14 Sortino and -23.93% drawdown. The result remains favorable across CPCV split distributions, but simulation shows drawdown risk is still path-dependent. Options are not automatically valuable. Fixed 10% SPY put and collar overlays only modestly change SPY's path after daily marks and hedge costs. When options are admitted inside the downside-score allocator, CVaR, volatility, maximum drawdown, and some simulated drawdown-breach rates improve, but realized Sortino is roughly unchanged and total return falls relative to the no-option book. The practical lesson is therefore not "buy options." It is: buy convexity only when the premium, skew, hedge cost, and portfolio drawdown states justify it.

1 Introduction

The investor in this paper begins with a familiar book: SPY for broad U.S. equity risk. The natural next question is not whether a larger asset list can produce a better in-sample efficient frontier. It almost always can. The harder question is whether the additions survive after they are treated as instruments that must be funded, hedged, rolled, and tested out of sample. The proposed universe is deliberately concrete: SPY, a subset of large index constituents, gold, copper, VIX futures, Treasuries, short-dated SPY option overlays, and explicitly documented diagnostics for omitted long-dated option proxies.

This is the right place to start because each proposed addition has a different economic job. The constituent basket can add concentration and cross-sectional equity exposure. Gold can hedge some inflation and real-rate states, but it is not a guaranteed crisis asset. Copper is procyclical and growth-sensitive. VIX futures can pay during volatility spikes, but they are expensive to hold through contango and roll. Treasuries can hedge equity weakness in some regimes and fail badly in inflation shocks. Options can create convexity, but the investor pays implied volatility, skew, spread, and sometimes hedge turnover before any crash protection is earned.

The paper's thesis is that cross-asset portfolio construction should be written as cashflow engineering under downside objectives. A portfolio is not a vector of weights attached to columns in a return matrix. It is a collection of domestic-currency cashflows, margin accounts, settlement rules, and contingent payoffs. A futures return is not an equity return with a different ticker. A VIX futures sleeve is not spot VIX. A protective put is not a free drawdown reducer. A Treasury hedge is not a scalar beta. The instrument must be built before it is optimized.

The objective is also part of the thesis. Mean-variance optimization is useful when the investor is willing to treat upside and downside variation symmetrically and when the mean and covariance estimates are strong enough. The investor here is different. The concern is upside-adjusted performance: earn a reasonable total return, avoid deep drawdowns, survive left-tail regimes, and use options only when convexity improves the whole book. Sortino ratio, lower partial moments, drawdown, CVaR, and stress-window returns therefore lead the paper. Sharpe is reported because it is informative, not because it is the investment target.

The running example is used throughout. At the accounting level, SPY and the constituent basket are paid equity exposures; gold and copper are futures or ETF proxies with roll and liquidity caveats; VIX exposure is represented by a tradable VIX-futures proxy rather than the non-tradable index; Treasuries enter through TLT and a rates-futures diagnostic; SPY options enter as 95% put and 105% call target overlays built from local OPRA month-end slices. At the objective level, each asset is admitted only if it improves the portfolio after costs and validation. The short-dated SPY put strikes are below spot and the collar call strikes are above spot; because the code selects the nearest listed strike, a leg can be near-ATM even when the target is 95/105. At the empirical level, the out-of-sample test asks whether these instruments improve SPY's Sortino ratio, drawdown, and tail loss.

The result is intentionally mixed. The multi-asset construction works better than the equity-only benchmark over the tested path. The no-option downside-score book has higher annual return, higher Sortino, and lower maximum drawdown than SPY. The equal multi-asset portfolio improves Sortino and CVaR but not maximum drawdown, which is a useful warning against overcrediting diversification by label alone. Fixed 10% SPY put and collar overlays modestly change SPY's path and slightly improve some drawdown and tail metrics, but they do not transform the exposure. Options help only conditionally: the option-enabled downside-score book lowers volatility, CVaR, maximum drawdown, and some simulated drawdown-breach rates, but

it gives up annual return and no longer improves realized Sortino.

That finding is the point of the paper. Options are not rejected because their average standalone return is often negative, and they are not accepted because their payoff diagram looks convex. They are evaluated by portfolio contribution under a physical probability measure after paying a risk-neutral or market-implied premium. Buying convexity is rational when the premium is low enough, the payoff arrives in the book's bad states, and the improvement in drawdown or downside loss is worth the carry. It is not rational merely because a backtest can draw a smoother line after the fact.

The paper proceeds as follows. Section 2 states the literature and research design. Section 3 defines the data and claim contract. Sections 4 and 5 build domestic cashflow accounting and instrument conventions. Sections 6 through 8 define the downside objective, optimizer menu, and hedge layer. Sections 9 and 10 describe the walk-forward, CPCV, simulation, and ML validation evidence. Sections 11 and 12 audit the claims and state the production gap. The appendices hold proof sketches, solver reductions, instrument tables, validation details, algorithms, implementation contracts, and the source ledger.

2 Related Literature and Research Design

The design starts from portfolio optimization but does not let the optimizer lead the story. Markowitz mean-variance theory gives the classical conversion from expected returns and covariance into weights [1]. The practical caveat is that the mean vector is rarely strong enough to deserve that authority. When alpha estimates are weak, simple sizing or risk parity is often the more honest object. When MVO is used, it has to be paired with shrinkage, constraints, costs, and validation. When parameter uncertainty is material, robust optimization is the safer language than pretending the plug-in inputs are known [15, 16, 17].

The risk objective follows the same discipline. Downside risk, CVaR, and drawdown are not afterthoughts in this paper; they are the main objects. Lower partial moments, CVaR, maximum drawdown, and CDaR are established portfolio construction criteria [3, 4, 21]. Sharpe remains useful for comparison and overfit diagnostics, but it is not the target that decides admission.

The derivative layer follows the standard split between pricing and allocation. Options are marked with no-arbitrage logic, BSM-style benchmarks, implied surfaces, and Greeks [5, 6, 11]. Once continuous frictionless hedging is relaxed, hedge quality and transaction costs become part of the economics rather than a technical nuisance [12, 11]. Futures, forwards, FX, and rates require stochastic discounting and numeraire discipline [7, 9, 10]. The mark is a pricing object. The allocation alpha is a physical-measure forecast after carry, risk premia, cost, and capacity.

Machine learning enters only as an estimator. It may estimate expected returns, regimes, volatility states, surface residuals, or execution costs. It does not bypass the portfolio problem. The finance-specific validation literature matters because overlapping labels, serial dependence, and repeated trials can make ordinary cross-validation misleading [13]. In this paper, ML scores widen or narrow a forecast distribution; they do not create a free admission rule.

The research design is therefore simple. Each claim is either standard source material, an adapted modeling choice, an empirical result from the reproducible code, or an original paper contribution. The body cites sources in the normal way and keeps empirical claims sample-specific unless the evidence supports a broader statement.

2.1 What is borrowed and what is new

The paper borrows the standard objects: mean-variance optimization, lower partial moments, CVaR, drawdown, robust uncertainty sets, risk-neutral valuation, Greeks, stochastic discounting, forward measures, and purged validation. It does not relabel those objects as new. The new contribution is the way they are wired together for admission. Every sleeve must pass through the same sequence:

contract → domestic cashflow → mark → forecast → friction → downside path test.

That sequence is more important than any single optimizer.

The closest literature usually handles one layer at a time. Portfolio books discuss allocation under risk and constraints. Derivatives texts discuss marks, Greeks, and hedges. Rates texts discuss numeraires and term-structure models. Machine-learning texts discuss validation and feature leakage. Backtesting texts discuss costs and capacity. A production book needs all of them at once. This paper’s scope is the integration layer: a format in which each asset class can be compared without hiding its financing and hedge economics.

That claim hierarchy is part of the economics, not just an editing convention. A sleeve that looks attractive in isolation can still fail because the premium is too high, the downside covariance is wrong, the hedge is too costly, the roll is mispriced, or the validation dispersion is too wide. If a sleeve is admitted, the paper has to show which states pay for it. If it is rejected, the rejection should be traceable to a specific economic or empirical failure rather than to vague optimizer preference.

3 Data and Public Research Contract

The paper has two contracts. The first is a theoretical contract: every instrument must be described by a payoff, mark, funding rule, margin or premium convention, exposure map, and friction profile. The second is an empirical contract: every number in the result section must be regenerated from files inside this sub-repo. The base run therefore uses a derived monthly training panel and a daily execution panel under data/processed/. The build scripts can recreate those panels from the broader repository data while this sub-repo remains inside the monorepo, but the paper itself runs from the committed and generated local processed files.

Object	Implementation
Sample	2015-02-28 to 2026-05-31
Core observations	136
Expected calendar months	136
Missing core month-ends	0
Option overlay months	127
Equity index	SPY from local adjusted-close cache
Stock subset	AAPL, MSFT, NVDA, AMZN, GOOGL, JPM, XOM, UNH equal-weight return
Gold/copper	GC=F and HG=F local continuous futures
Volatility	Nearest VIX future proxy, not spot VIX
Rates	TLT proxy; ZN=F retained in panel
Options	SPY nearest-listed 95% put / 105% call and 95/105 collar targets

Table 1: Empirical data contract for the running SPY-centered example.

The data choices are deliberately practical. SPY is the equity benchmark. The stock sleeve is an equal-weight basket of AAPL, MSFT, NVDA, AMZN, GOOGL, JPM, XOM, and UNH. This is not a point-in-time full index replication; it is a representative liquid subset used to show how constituent risk enters the allocator. Gold and copper use local continuous futures series. VIX exposure is a nearest VIX-futures proxy because spot VIX is not tradable. Rates use TLT, with

ZN futures retained in the processed panel as a diagnostic. SPY options use month-end OPRA slice trade closes to construct nearest-listed 95% put and 105% call target overlays. The code also builds a diagnostic constituent-option file that selects up to 100 point-in-time S&P members by trailing dollar volume and marks modeled two-year ATM calls. Because the local data do not contain complete historical two-year listed chains for every selected stock, that diagnostic is excluded from the headline optimizer, figures, and result tables.

The processed sample runs from February 2015 through May 2026 with no missing core month-ends after the timing repair. The rolling experiment starts in February 2018 after a 36-month training window. The final month is excluded whenever the local price data end before month-end. This matters in the current repository because daily price caches reach June 2026, but June is incomplete. The paper reports the complete February 2018–May 2026 out-of-sample period.

Instrument	Class	Accounting	Main friction
SPY EQ_BASKET	equity ETF large-cap stocks	paid spot exposure plus distributions paid spot basket return	spread, impact, cash financing survivorship, corporate actions, single-name liquidity
GC_FUT	gold future proxy	variation-margin return from continuous proxy	roll, margin, delivery-calendar simplification
HG_FUT	copper future proxy	variation-margin return from continuous proxy	roll, margin, cyclicalities
VX_FRONT	VIX future proxy	front-contract futures return	roll drag, convex stress behavior, non-spot tradability
TLT SPY_95P	Treasury ETF proxy listed option overlay	paid duration exposure premium paid, expiry payoff or sale/exercise value	DV01, financing, rates-regime risk bid/ask, skew, theta, quote timing
SPY_95_105_COLLAR	listed option overlay	long put plus short call terminal cashflow	assignment, capped upside, bid/ask

Table 2: Instrument normalization contract generated by the local package.

The empirical results remain research proxies. Continuous futures are price-spliced; roll yield is approximated through a deterministic roll ledger where explicit contract metadata are available. The VIX-futures series uses a predeclared front-contract roll rule, not a production roll engine. SPY option overlays use monthly slice closes, exact listed expiry dates, raw SPY expiry prices, and daily model marks between quote anchors; they are not a full intraday execution simulator. The omitted top-100 ATM call diagnostic uses point-in-time membership and price history, but modeled option marks. The stock basket has survivorship and selection limits. These limits do not invalidate the experiment, but they determine what can be claimed from it. The paper can test whether the framework behaves sensibly on real local data. It cannot claim deployable trading alpha.

Every object used at review date t_k must be available at t_k . The outer allocation forms monthly targets from a trailing 36-month window ending at the last completed month-end. The inner execution layer applies each target after the configured one-trading-day lag and records daily sleeve P&L, costs, rolls, hedge trades, and drawdown-control actions. For SPY options, the trade log records the exact listed expiry as the economic return date. The monthly panel separately buckets that realized option return into the month-end containing the expiry. The terminal SPY price is the expiry close or the last available close before expiry. The option terminal rule compares sale proceeds from any available post-entry quote at or before expiry with exercise value; when no such quote exists, the option is exercised if in the money and otherwise expires. Months without a valid option slice keep the core asset panel and set option-sleeve returns to zero. The omitted top-100 ATM call diagnostic selects constituents and volatility inputs using only data available at the prior month-end and marks the daily LEAPS-style roll before the next rebalance. Risk-free rates use the local $\wedge$ IRX series and are used only in metric calculation. No result uses future

returns to set weights.

The computed tables and figures are regenerated from the local pipeline rather than edited by hand. That is a mechanical constraint, but it has a substantive purpose: it keeps the result section from drifting away from the code that produced it. The same discipline applies to claim strength. Definitions are conventions, literature claims require citations, empirical claims require generated artifacts, and production claims would require executable data, broker costs, and paper-trading parity. This paper makes no production claim.

4 Domestic Cashflow Accounting

The accounting numeraire is USD cash. Let B_t^d be the domestic money-market account and let $\mathcal{F}_t$ contain only information available at time t . A tradable instrument is represented by a payoff map, a marking map, an exposure map, funding and margin rules, settlement rules, and a friction profile.

Definition 1 (Instrument cashflow object). *An instrument i is the tuple*

$$\mathbf{i} = (\Pi_i, P_i, H_i, F_i, M_i, S_i, C_i),$$

where Π_i is the contractual payoff map, P_i is the mark, H_i is the exposure map, F_i is the funding convention, M_i is the margin convention, S_i is the settlement rule, and C_i is the friction profile.

Result 1 (Cashflow normalization). *If an instrument has finite domestic value, a specified settlement rule, and a specified friction profile, its one-period contribution can be written as a domestic discounted cashflow increment minus implementation charges:*

$$\Delta \widetilde{W}_i = \frac{\Pi_i(Y_{\leq t+h})}{B_{t+h}^d} - \frac{P_i(t, Y_t)}{B_t^d} - \widetilde{C}_i(q_i, q_{i,-}, Y_{\leq t+h}).$$

The cross-asset portfolio cashflow is the sum of these terms across instruments and cash accounts. A reported return divides the aggregate discounted cashflow by the capital, premium budget, margin allocation, or other denominator specified by the mandate.

The point is basic but important. Once everything is expressed as domestic cashflows, the optimizer can compare sleeves without pretending their mechanics are the same. An equity position uses paid capital and earns dividends. A futures position uses variation margin and roll discipline. An FX forward exchanges currencies under collateral and funding rules. A bond is a strip of dated cashflows with duration and convexity. An option is a premium plus a nonlinear payoff or mark, with hedge error and Greek exposure.

Result 2 (Pricing measure is not allocation measure). *Let N_t be a positive tradable numeraire. A payoff can be marked under the measure that makes prices divided by N_t martingales:*

$$P_t = N_t \mathbb{E}_{\mathbb{Q}^N} \left[\frac{\Pi_T}{N_T} \mid \mathcal{F}_t \right].$$

The allocation decision then uses the physical-measure forecast of the marked domestic return after premia, forecast uncertainty, costs, and capacity. The risk-neutral drift that marks a claim is not the expected return that admits it to a portfolio.

This is the main guardrail for FX and rates. A foreign bond, a currency forward, a caplet, or a Treasury future must first be marked consistently. Only after that step can a real-world carry or risk-premium forecast enter the allocator.

4.1 Return decomposition

For a review horizon h , the portfolio return is decomposed as

$$R_{k+1}(x; x_-) = \text{carry} + \text{spot move} + \text{curve move} + \text{vol move} + \text{roll} + \text{convexity} + \text{theta} \\ - \text{spread} - \text{impact} - \text{funding} - \text{margin drag} - \text{hedge cost} - \text{data penalty}.$$

The decomposition is not meant to force every instrument to have every term. It is a checklist. An equity may have no roll term. A futures sleeve has roll and margin. An option has theta, convexity, and hedge cost. A bond has carry, roll-down, DV01, and convexity. An FX forward has spot currency movement plus forward points and funding.

The allocator sees the net quantity:

$$R_{k+1}(x; x_-) = x^\top X_{k+1} - C^{\text{entry}}(x - x_-) - C^{\text{hold}}(x) - C^{\text{hedge}}(x) - C^{\text{model}}(x).$$

Here X_{k+1} is generated by repricing or scenario cashflows, not only by a covariance matrix. The cost terms are allowed to be nonlinear, which is why the feasible payoff set is not the classical linear span.

4.2 Friction-deflated attainable set

Let $\mathcal{U}$ be the instrument universe, let $\mathcal{K}(\mathcal{U})$ be a convex feasible set, and let $G(x)$ be the gross statewise payoff from positions x in a common domestic numeraire. The friction-deflated attainable set is

$$\mathcal{A}_\phi(\mathcal{U}) = \{G(x) - C_\phi(x) - m : x \in \mathcal{K}(\mathcal{U}), m \geq 0\}.$$

The free-disposal term m records the fact that an investor can always hold extra cash or discard surplus payoff in the ordering used for risk constraints. Without frictions, spans are linear cones. With convex spreads, impact, hedge bands, funding, and capacity costs, the free-disposal closure is convex under the stated assumptions but generally not conic. Doubling a position can more than double costs. That geometry is the reason admission is not automatic when a new instrument appears in the data set.

Remark 1. *The paper does not require every production cost to be perfectly known. It requires every material cost to be either modeled, stressed, or listed as a production gap. Unknown frictions are not zero; they are unresolved.*

5 Instrument Conventions

The framework uses the same accounting object for every asset class, but each class keeps its own convention. Equities and ETFs are spot positions with dividends, borrow, corporate actions, spread, impact, and capacity. Listed options are premium-paid nonlinear claims with delta, gamma, vega, theta, exercise style, surface marks, and hedge costs. Futures are notional contracts with daily variation margin, roll calendars, delivery avoidance, and margin drag. Forwards are zero-upfront bilateral exchanges whose value depends on funding, collateral, and stochastic discounting. FX positions carry domestic and foreign cash accounts. Fixed income is represented by zero-coupon or coupon cashflows, DV01, key-rate DV01, roll-down, convexity, and financing.

Instrument	Linear exposure	Convexity	Vol/rates exposure	Data risk	Funding
SPY	equity beta	none	none	low	paid cash
EQ_BASKET	single-name beta	none	none	medium	paid cash
GC_FUT	commodity delta	none	none	medium	variation margin
HG_FUT	commodity delta	none	none	medium	variation margin
VX_FRONT	volatility-futures delta	roll convexity	vol regime	high	variation margin
TLT	duration beta	bond convexity	key-rate proxy	medium	paid cash
SPY_95P	negative equity delta	gamma	vega/theta	high	premium
SPY_95_105_COLLAR	reduced equity delta	put/call gamma	skew/theta	high	premium net short-call risk

Table 3: Exposure map used by the allocator.

The convention table prevents a common error. A futures sleeve may have attractive trend or carry behavior but still fail because of roll cost, delivery constraints, or margin use. An option overlay may improve a payoff diagram but fail once gamma-driven hedge turnover is charged. A duration hedge may have unattractive standalone expected return but still improve the book if it pays in drawdown states. The instrument is admitted for what it does to the funded portfolio path, not for what it looks like in isolation.

Equities are the benchmark case. A long position pays price appreciation and dividends, uses paid capital, and incurs spread, commission, impact, and possibly borrow if short. The main modeling risks are survivorship bias, corporate-action handling, stale universe membership, and liquidity. If the accounting machinery cannot handle paid spot positions cleanly, it should not be trusted with derivatives.

Options change the problem because the premium is paid today and the payoff arrives only in selected states. The mark comes from an option-pricing map or market quote. The risk map includes at least delta, gamma, vega, theta, and exercise style. The allocation forecast can come from volatility risk premium, skew, surface residuals, event risk, or hedge value, but the cost model must include bid/ask, model error, and hedge cost. For American or hard-to-borrow underlyings, early exercise and borrow assumptions must be explicit. For thin options, quote filtering and stale marks are part of the data-quality charge.

Futures and forwards are different again. A futures position is notional exposure with daily variation margin; treating it as an equity weight hides the financing convention. The strategy must define the contract chain, roll date, first-notice or last-trade constraints, margin rate, roll spread, and delivery avoidance. A forward is a zero-upfront terminal exchange under collateral and funding assumptions. FX positions require domestic and foreign money-market accounts. A carry signal is not free yield; it is compensation for a funded currency position with spot risk and roll risk.

The first version uses rates instruments to enforce the no-arbitrage and duration accounting discipline, not to replace a rates library. It covers zero-coupon bonds, coupon-bond proxies, DV01, key-rate DV01, Treasury futures, and roll-down. A production rates book would add collateral discounting, benchmark reform details, swap curves, repo, CTD optionality, futures basis, accrued interest, coupon selection, and tax/accounting rules.

6 Objective Hierarchy

The primary objective is robust Sortino improvement [2, 21]. For a target return τ , portfolio return $R(x)$, and ambiguity set $\mathcal{U}$, the decision problem is

$$\max_x \inf_{u \in \mathcal{U}} \frac{\mathbb{E}_u[R(x)] - \tau}{\sqrt{\mathbb{E}_u[(\tau - R(x))_+^2]}}$$

subject to budget, gross, liquidity, margin, stress, and turnover constraints, with the denominator required to be positive or floored in numerical implementations. Sharpe is reported because readers know it and because deflated-Sharpe diagnostics help measure overfit. It does not decide the portfolio.

Tail and path constraints sit beside Sortino. CVaR controls scenario losses:

$$\text{CVaR}_\alpha(L) = \min_{\eta} \left\{ \eta + \frac{1}{1-\alpha} \mathbb{E}[(L - \eta)_+] \right\}.$$

Drawdown is pathwise. For wealth $W_t > 0$, running peak $P_t = \max_{s \leq t} W_s$, and relative drawdown $d_t = W_t/P_t - 1 \leq 0$, maximum drawdown controls $\min_t d_t$, while CDaR controls the tail average of drawdown magnitudes. These constraints matter for hedges because a sleeve can be valuable even when its standalone mean is negative.

Proposition 1 (Drawdown hedge admission). *A hedge with negative standalone mean can be optimal if its payoff is sufficiently positive in the portfolio states that set the running peak-to-trough losses. In that case the hedge is not admitted as alpha; it is admitted as path insurance.*

Capacity is part of the objective hierarchy. If gross profit and spread cost scale roughly linearly with capital but impact scales superlinearly, the relevant output is not one return number. It is a capacity curve showing where marginal alpha is consumed by implementation.

For target τ and order p , the lower partial moment is

$$\text{LPM}_p(R; \tau) = \mathbb{E}[(\tau - R)_+^p].$$

The Sortino denominator is the square root of the second lower partial moment. When the sample shortfall is zero, the ratio is not a meaningful finite objective; code should report it as undefined or use a documented numerical floor for search. With that convention in place, Sortino is the relevant object when upside volatility is not the problem. A long-convexity sleeve can increase total volatility and still improve Sortino if it pays below the target. A short-volatility sleeve can increase average return and still fail because the losses sit exactly in the downside states.

In scenarios $s = 1, \dots, S$ with losses $L_s(x)$, the CVaR constraint can be written with auxiliary variable η and slacks u_s :

$$\begin{aligned} \eta + \frac{1}{(1-\alpha)S} \sum_{s=1}^S u_s &\leq \bar{c}, \\ u_s &\geq L_s(x) - \eta, \quad u_s \geq 0. \end{aligned}$$

This representation is useful because it can be embedded in linear or convex programs once the loss map is convex. It also gives a clean stress interpretation: the optimizer must pay for the average of the bad tail, not only the single worst point.

For scenario path j and time t , let $W_{j,t}$ be wealth, $P_{j,t}$ the running peak, and $D_{j,t}$ the positive relative drawdown magnitude:

$$P_{j,t} \geq P_{j,t-1}, \quad P_{j,t} \geq W_{j,t}, \quad D_{j,t} = 1 - \frac{W_{j,t}}{P_{j,t}}.$$

Maximum drawdown constrains $\max_{j,t} D_{j,t}$. CDaR applies a CVaR construction to the drawdown samples. This is heavier than variance, but it captures the reason investors buy hedges: survival across the path.

7 Choosing the Optimizer

The method is chosen by input quality. If expected returns are weak or unstable, the default is equal weight, inverse volatility, risk parity, or hierarchical risk parity. If forecasts are explicit and validated, robust MVO, robust Sortino, CVaR, or CDaR optimization can be used. If views are subjective, they enter as priors with uncertainty. If machine learning is used, the output is a forecast distribution with validation error, not a direct trade.

Method	Use when	Current implementation	Main failure
Equal multi	alpha is weak	equal weights in six core sleeves	ignores expected return and state payoff
Inverse vol	risk estimates are usable but means are weak	rolling 36-month inverse-vol weights	can overweight low-vol drawdown assets
Downside score	Sortino/drawdown path is the mandate	candidate search with costs, caps, turnover	heuristic, not global proof
ML downside score	lagged forecasts survive purged CV	forecast-to-weight admission with frozen model	fold IC may not translate to portfolio value
CVaR/CDaR program	tail or path loss is explicitly capped	mathematical reduction in appendix	heavier solver and scenario-design burden

Table 4: Allocation methods and when they are allowed to lead the decision.

Result 3 (Sortino admission). *At a current portfolio x , a small sleeve z should enter a downside-oriented book only if its robust net mean clears the downside-state covariance hurdle and its entry cost. In words, the sleeve must pay for the states in which the investor is short of the target return, not only increase average return in calm states.*

This admission rule explains why simple methods remain in the menu. Optimizers maximize estimation error when their inputs are not strong enough. A simple allocation that survives costs, drawdowns, and validation can be more useful than a sophisticated allocation driven by a fragile mean vector.

Equal weight is a data-scarce benchmark. Inverse volatility adds a minimal risk estimate. Risk parity tries to equalize risk contributions. Hierarchical risk parity uses correlation structure without requiring a mean vector. These rules are not naive in the bad sense. They are honest when the alpha estimate is weak. Their main limitation is that they cannot express state-contingent payoff quality unless the instrument universe has already been engineered well.

The robust MVO version used in the code can be written as

$$\max_x (\hat{\mu} - \delta_\mu)^\top x - \frac{\lambda}{2} x^\top (\hat{\Sigma} + \Delta_\Sigma) x - C_\phi(x)$$

subject to budget, margin, gross, and capacity constraints. The haircut terms are not decorative. They are the mechanism that prevents a weak or unstable forecast from dominating the allocation.

Sortino optimization is usually less convenient than MVO because it is a ratio with a downside denominator. In production it can be solved by grid search for small universes, fractional programming, smooth approximations, or scenario-based nonlinear solvers. CDaR optimization uses path variables and can be heavier still. The paper does not require one solver to dominate. It requires the chosen solver to match the risk question.

Views enter only when the prior, view vector, confidence, and horizon are explicit. A view such as “duration should hedge the equity book in recession states” is not the same as a point forecast of bond return. It should be encoded as a scenario or state-conditional view with uncertainty. The allocator then decides whether the view survives costs and downside constraints.

8 Dynamic Hedge Layer

The outer allocation is monthly in the base implementation. The hedge layer is faster. Delta, DV01, key-rate, FX, futures, vega, and gamma exposures are measured from the current book and reduced only when the expected benefit exceeds the trading cost. Continuous frictionless hedging is not assumed.

Result 4 (Hedge cost and no-trade bands). *For an option position with gamma exposure, proportional spread, and realized volatility, the hedge problem is an inner impulse-control problem. Wider spreads widen the no-trade band; larger gamma and higher realized volatility narrow it. The outer allocator should therefore see a superlinear hedge-cost charge rather than a free delta hedge.*

This result is why option sleeves are treated carefully. A covered-call sleeve may look attractive before hedge costs and assignment risk. A protective-put sleeve may look too expensive by standalone carry but improve drawdown. A short-vol sleeve may pass in calm states and fail in jump states. The hedge layer turns those tradeoffs into explicit costs and risk limits.

Rates and futures hedges follow the same logic. A Treasury future can reduce key-rate risk, but it brings CTD basis, roll timing, margin, and delivery avoidance. An FX hedge can remove spot currency risk, but it creates forward-roll and funding exposure. The hedge is part of the portfolio, not an external cleaning step.

8.1 Delta and gamma

For option value $V(S, t, \sigma)$, delta and gamma are

$$\Delta = \frac{\partial V}{\partial S}, \quad \Gamma = \frac{\partial^2 V}{\partial S^2}.$$

A delta hedge removes first-order spot exposure locally. It does not remove gamma, vega, jump, liquidity, or discrete-rebalancing risk. If the underlying moves through the no-trade band, the hedge is reset and the cost is charged to the book. If the option is short gamma, realized volatility increases turnover and losses. If the option is long gamma, the hedge may monetize movement but still pays spread and impact.

8.2 DV01 and key-rate hedging

For fixed income, the local exposure is not a single beta. A portfolio can be neutral to parallel duration and still exposed to slope or curvature. The exposure vector is

$$h^{rates}(x) = (\text{KRD}_{2Y}(x), \text{KRD}_{5Y}(x), \text{KRD}_{10Y}(x), \dots).$$

Hedging this vector with Treasury futures or swaps introduces basis and roll. A hedge that looks clean in key-rate space can still fail if the chosen instrument's CTD bond, coupon, or repo specialness changes.

8.3 FX and futures rolls

FX hedges are rolled because forwards mature. Futures hedges are rolled because contracts expire and physically settled contracts may enter delivery windows. A roll rule must be defined before the backtest starts. A strategy that chooses roll dates after seeing the path is not a strategy; it is a fitted execution assumption. The base implementation treats roll cost as part of the friction profile and requires delivery avoidance as a hard constraint.

9 Empirical Workflow

The empirical workflow is built to make the theory answerable. The question is not whether one optimized line beats SPY in one realized path. The question is whether the proposed cashflow-and-drawdown framework survives several screens that are hard to pass for different reasons. The realized path shows what actually happened from February 2018 through May 2026. Combinatorial purged cross-validation asks whether the fitted rule depends on a particular ordering of held-out months. Monte Carlo asks how fragile the drawdown result is to alternative paths with the same empirical building blocks. The ML arm asks whether a forecast layer adds information after purging, embargo, fold dispersion, and cost stress.

9.1 Data construction

The reproducible sample now has two layers. The strategic layer is monthly because the paper studies portfolio construction rather than intraday trading. The execution layer is daily because costs, option marks, futures rolls, hedge decisions, and drawdown controls are not monthly abstractions in a live book. SPY and TLT use adjusted closes. The stock sleeve is an equal-weight basket of available large-cap constituents, averaged at the daily-return level before monthly aggregation. Gold and copper use local continuous futures proxies. VIX exposure is represented by a contract-specific front VIX-futures proxy rather than spot VIX, because spot VIX is not directly tradable.

SPY option overlays are selected from month-end OPRA slices: a nearest-listed 95% put target and a 95/105 collar target. The option payoff calculation uses raw SPY expiry prices so that strikes and terminal underlying prices share the same units. The daily execution panel then carries those option positions from entry to expiry. Entry and terminal states are quote or expiry anchored; interim daily marks are model marks when matched daily OPRA quotes are unavailable. A separate long-dated constituent-option diagnostic is built and audited, but it is not part of the headline research universe because the local cache does not contain complete historical listed LEAPS chains.

The panel is a research fixture, not a claim of institutional execution quality. The commodity series remain continuous proxies. The VIX-futures sleeve has contract-specific settlement metadata but not a full production execution book. The stock basket is representative rather than a full point-in-time S&P 500 reconstruction. Those limitations are not hidden; they are why the evidence is treated as a research screen.

9.2 Realized out-of-sample path

The historical experiment uses a 36-month rolling training window. At each monthly decision date, the rule sees only the trailing completed window and chooses strategic sleeve targets. The daily execution layer applies the latest known target on later trading dates, pays turnover costs, tracks option lifecycle marks, applies a lagged delta-hedge rule with a no-trade band, records VIX-futures roll events, and writes a separate drawdown-control variant rather than hiding risk cuts inside every strategy. The headline table reports monthly returns aggregated from that daily executed P&L. CPCV still validates the monthly allocation rule; Monte Carlo uses the daily executed return paths when they are available.

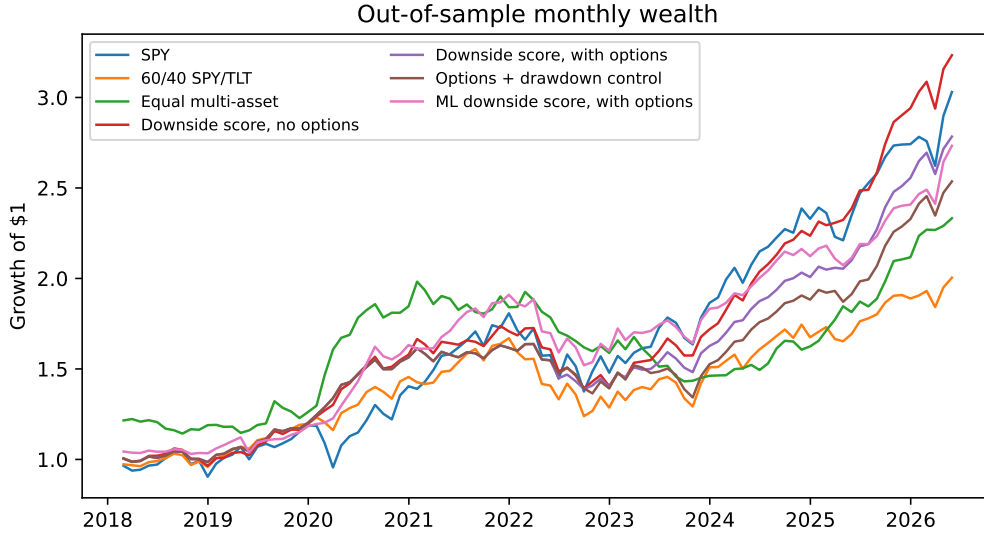


Figure 1: Out-of-sample monthly wealth from February 2018 through May 2026.

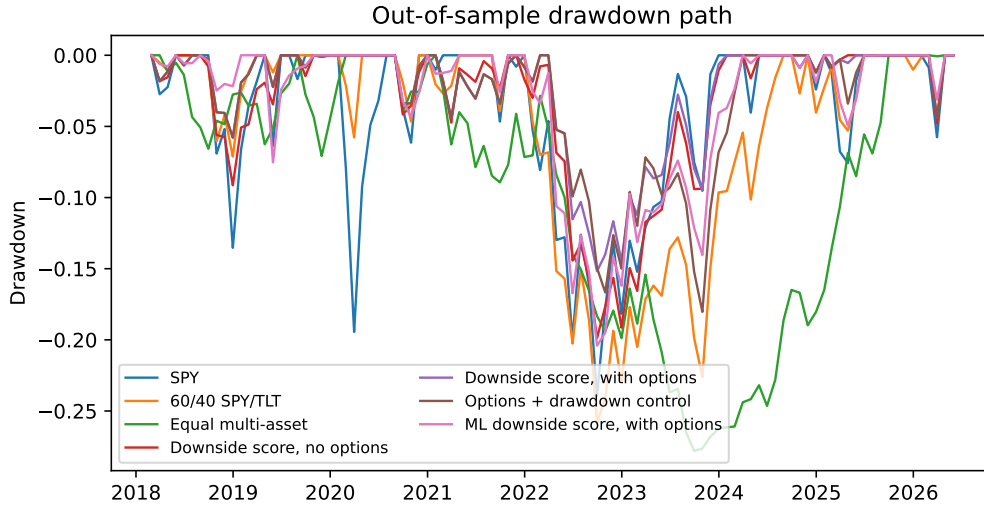


Figure 2: Out-of-sample drawdown paths. The figure is the path test behind the Sortino result.

9.3 Portfolio rules

The benchmarks are deliberately simple. SPY measures the starting book. The 60/40 benchmark uses 60% SPY and 40% TLT. The equal multi-asset benchmark holds SPY, the constituent basket, gold, copper, VIX futures, and TLT in equal weights. Rolling inverse volatility estimates trailing volatility and scales the same six non-option assets.

The two non-ML optimized books use the same candidate-search engine and the same training window. The no-option downside-score book can hold SPY, the constituent basket, gold, copper, VIX futures, and TLT. The option-enabled downside-score book can also hold the standalone SPY 95% put return and the 95/105 collar return, each capped as an option-notional sleeve. The score combines trailing Sortino, annualized return, drawdown, volatility, concentration, and turnover cost. This is a transparent research proxy for the objective hierarchy, not a claim that random candidate search is the final production optimizer.

Strategy	Average OOS weights	Option notional	Monthly turnover
Equal multi-asset	SPY:17%, EQ_BASKET:17%, GC_FUT:17%, HG_FUT:17%, VX_FRONT:17%, TLT:17%	0.0%	0.0%
ML downside score, no options	SPY:36%, EQ_BASKET:19%, GC_FUT:13%, HG_FUT:11%, VX_FRONT:1%, TLT:19%	0.0%	39.8%
ML downside score, with options	SPY:30%, EQ_BASKET:15%, GC_FUT:11%, HG_FUT:11%, VX_FRONT:2%, TLT:17%, SPY_95P:7%, SPY_95_105_COLLAR:7%	13.2%	36.1%
Rolling inverse vol	SPY:20%, EQ_BASKET:16%, GC_FUT:23%, HG_FUT:15%, VX_FRONT:4%, TLT:23%	0.0%	3.0%
60/40 SPY/TLT	SPY:60%, TLT:40%	0.0%	0.0%
Downside score, no options	SPY:24%, EQ_BASKET:31%, GC_FUT:18%, HG_FUT:9%, VX_FRONT:6%, TLT:12%	0.0%	15.0%
Downside score, with options	SPY:18%, EQ_BASKET:29%, GC_FUT:14%, HG_FUT:9%, VX_FRONT:5%, TLT:9%, SPY_95P:7%, SPY_95_105_COLLAR:9%	15.8%	15.7%
Options + drawdown control	SPY:18%, EQ_BASKET:29%, GC_FUT:14%, HG_FUT:9%, VX_FRONT:5%, TLT:9%, SPY_95P:7%, SPY_95_105_COLLAR:9%	15.8%	15.7%
SPY	SPY:100%	0.0%	0.0%
SPY + 10% collar	SPY:100%, SPY_95_105_COLLAR:10%	10.0%	20.0%
SPY + 10% put	SPY:100%, SPY_95P:10%	10.0%	10.0%

Table 5: Average out-of-sample weights for the rolling allocation rules.

Strategy	Months	Legacy total	Daily-executed total	Mean diff.
SPY	100	202.69%	203.00%	0.00%
Downside score, no options	100	193.01%	223.36%	0.10%
Downside score, with options	100	143.35%	178.40%	0.14%
Options + drawdown control	100	143.35%	153.62%	0.05%

Table 6: Legacy monthly proxy returns versus monthly returns aggregated from daily execution.

Option sleeve	Mark source	Rows
SPY_95P	expiry_terminal_value	127
SPY_95P	model_mark_from_entry_iv	1537
SPY_95_105_COLLAR	expiry_terminal_value	127
SPY_95_105_COLLAR	model_mark_from_entry_iv	1537

Table 7: Option mark provenance in the daily execution layer.

Strategy	Hedge rows	Hedge trade cost	Hedge P&L
Downside score, with options	1324	0.08%	7.93%
Options + drawdown control	1323	0.08%	7.43%

Table 8: Daily delta-hedge ledger summary. Hedge decisions use prior-day mark-ledger deltas or fixed proxy deltas.

Strategy	Trades	Avg. turnover	Total cost
Downside score, no options	376	6.5%	0.93%
Downside score, with options	336	7.2%	2.67%
Options + drawdown control	345	11.9%	3.62%

Table 9: Daily execution cost attribution for target rebalances.

Date	VX contract	Settlement	Rule
2013-01-02	VXF13	2013-01-16	nearest_settlement_delivery_avoidance
2013-01-15	VXG13	2013-02-13	nearest_settlement_delivery_avoidance
2013-02-12	VXH13	2013-03-20	nearest_settlement_delivery_avoidance
2013-03-19	VXJ13	2013-04-17	nearest_settlement_delivery_avoidance
2013-04-16	VXK13	2013-05-22	nearest_settlement_delivery_avoidance
2013-05-21	VXM13	2013-06-19	nearest_settlement_delivery_avoidance
2013-06-18	VXN13	2013-07-17	nearest_settlement_delivery_avoidance
2013-07-16	VXQ13	2013-08-21	nearest_settlement_delivery_avoidance

Table 10: Predeclared VIX-futures roll events in the daily execution panel.

Strategy	Triggers	Worst prior DD	Threshold
Options + drawdown control	22	-10.84%	-10.00%

Table 11: Drawdown-control variant. Signals are based on prior-close drawdown and applied after the trigger.

9.4 Option representation

The paper treats the option overlay as a portfolio instrument, not as a separate trade advertisement. A put bought at time t for premium $P_t(K, T)$ contributes

$$R_{t,T}^{put} = \frac{(K - S_T)_+ - P_t(K, T)}{S_t}.$$

This is a physical-measure realized contribution after paying the market premium. The premium itself is a risk-neutral or market-implied price; the expected contribution depends on the physical distribution of S_T , the value of payoff states to the portfolio, and implementation costs. The collar adds the short-call leg:

$$R_{t,T}^{collar} = R_{t,T}^{put} + \frac{C_t(K_c, T) - (S_T - K_c)_+}{S_t}.$$

The formulas make clear why options can improve drawdown and still reduce average return. The premium is paid in every state. The convex payoff arrives only in selected states. The rolled constituent call basket uses the one-month mark-to-market analog

$$R_{t,t+1}^{2yATM} = \frac{\max\{B_{t+1}(S_{t+1}, K, T - \Delta t), (S_{t+1} - K)_+\} - A_t(S_t, K, T)}{S_t},$$

$$K = S_t, \quad T = 2.$$

where A_t is the entry ask proxy, B_{t+1} is the exit bid proxy, and the maximum implements the sell-versus-exercise decision used by the proxy. In a listed American call, early exercise before expiry is usually dominated when the option retains time value and dividend/borrow effects are not decisive. The rule is therefore a conservative cashflow accounting convention rather than a claim that early exercise is normally optimal.

9.5 CPCV and overfit controls

The validation module implements purging, embargo, combinatorial purged cross-validation, and a probability-of-backtest-overfitting diagnostic. The default geometry uses four chronological groups and two held-out groups, giving six CPCV splits. For a monthly panel this is intentionally modest. It is not a substitute for a larger live history, but it forces each strategy to be refit and tested on alternative held-out histories instead of replaying the realized path.

The source distinction matters [13, 8]. Purging and embargo protect the train-test boundary when labels overlap. CPCV generates a distribution of out-of-sample paths. PBO asks whether the selected configuration looks good mainly because many alternatives were tried. None of these tools establishes tradable edge. They make the result harder to overstate.

9.6 Reality-preserving simulations

The simulation layer does not invent alpha. It resamples or simulates returns conditional on the observed strategy return process. Circular block bootstrap preserves short-run path dependence. IID empirical resampling is retained as a weaker comparison. The EGARCH arm uses a fitted EGARCH model when the daily executed sample is long enough and otherwise uses an EWMA residual fallback. In the current run, the simulation input is the daily executed strategy return path. The diagnostics table reports lag-1 autocorrelation, skew, excess kurtosis, and the recommended interpretation.

This matters because drawdown is a path statistic. A strategy can have the same average period return and still have a very different probability of breaching an 8%, 10%, or 15% drawdown threshold. The simulation tables are therefore treated as uncertainty screens, not as new empirical histories.

9.7 ML forecast arm

Machine learning enters only as a forecast estimator. The feature set is lagged: trailing returns, realized volatility, drawdown state, cross-asset correlation, VIX state, gold and copper momentum, rates trend, option-overlay payoff history, and simple regime flags. The baseline model is regularized linear regression. LightGBM is allowed in the validation table when installed, but hyperparameters are selected inside purged CV and are frozen before the realized walk-forward path is evaluated.

The admission rule is conservative. A model must show useful fold IC, acceptable fold dispersion, CPCV survival, tolerable PBO, and cost robustness before it can become a headline portfolio. If the ML arm produces a plausible-looking OOS line but fails those screens, it remains a documented rejection result. That is an empirical finding, not a software failure.

9.8 Stress and regime cuts

Stress windows and regime buckets are descriptive cuts, not optimizer inputs. The generated tables report realized strategy behavior in calm versus stress states and option overlay payoffs by VIX/regime bucket. The goal is to show which states pay for a sleeve. In this paper the short-dated SPY put and collar have their best upper-tail payoffs in stress states. That is why options must be valued at the portfolio level rather than by standalone average return or payoff shape alone.

10 Results

The evidence is strongest where the theory is simplest. A diversified cross-asset book and a downside objective improve the realized SPY path. The option conclusion is conditional, and the ML arm is not admitted as a headline improvement. This is a better result than a clean success story because it separates three effects: diversification, convexity, and forecasting.

10.1 Realized OOS path

Strategy	Ann. ret.	Ann. vol.	Sortino	Sharpe	Max DD	Calmar	CVaR95	Hit rate
SPY	14.23%	16.56%	1.14	0.73	-23.93%	0.59	9.51%	67.0%
SPY + 10% put	14.49%	16.56%	1.18	0.75	-23.12%	0.63	9.32%	67.0%
SPY + 10% collar	14.47%	16.55%	1.17	0.75	-23.31%	0.62	9.34%	67.0%
60/40 SPY/TLT	8.70%	12.31%	0.80	0.53	-25.82%	0.34	6.90%	66.0%
Equal multi-asset	10.70%	13.19%	1.46	0.64	-27.80%	0.38	4.04%	59.0%
Rolling inverse vol	12.48%	9.71%	1.73	1.00	-19.36%	0.64	5.12%	65.0%
Downside score, no options	15.12%	10.35%	2.00	1.18	-19.87%	0.76	5.66%	70.0%
Downside score, with options	13.07%	8.87%	1.99	1.15	-15.18%	0.86	4.63%	67.0%
Options + drawdown control	11.82%	9.73%	1.59	0.94	-18.04%	0.66	4.97%	65.0%
ML downside score, no options	13.55%	12.38%	1.41	0.89	-22.72%	0.60	7.22%	68.0%
ML downside score, with options	12.82%	10.88%	1.53	0.93	-20.41%	0.63	6.64%	69.0%

Table 12: Out-of-sample performance, February 2018–May 2026.

SPY earns a strong positive return over the sample, but it is not the best downside path. After daily execution, SPY earns 14.23% annualized with a 1.14 Sortino ratio and a -23.93% maximum drawdown. Equal multi-asset exposure no longer looks like a clean improvement: it earns 10.70% annualized, has a 1.46 Sortino, and draws down -27.80%. The rolling inverse-volatility benchmark is more useful, with 12.48% annualized return, a 1.73 Sortino, and a -19.36% drawdown. This distinction matters. Diversification is not valuable merely because more asset labels are present; the allocation rule must also avoid paying for sleeves that add drawdown or carry at the wrong time.

The no-option downside-score portfolio is the highest-return headline performer. It earns 15.12% annualized with 10.35% volatility, a 2.00 Sortino ratio, a -19.87% maximum drawdown, and 5.66% CVaR95. The average allocation is economically plausible: it keeps equity exposure, uses the stock basket for cross-sectional participation, holds gold and copper for real asset and growth states, carries a capped VIX-futures sleeve, and keeps duration modest through a sample that includes the 2022 rates shock.

The option-enabled downside-score portfolio is more defensive but not better on total return. It earns 13.07% annualized with 8.87% volatility, a 1.99 Sortino ratio, and a -15.18% maximum drawdown. CVaR95 improves from 5.66% in the no-option book to 4.63%. The cost is lower annual return. That tradeoff is the right way to read the option result: options reduce several downside-loss measures, but they do not dominate the simpler book once total return and realized Sortino are both part of the objective. The explicit drawdown-control variant is also not free. It earns 11.82% annualized with a 1.59 Sortino and a -18.04% drawdown, so the rule reduces some risk states but gives up too much upside in this sample.

The fixed SPY overlays are a useful modest-result check. Adding a 10% 95% put overlay to SPY raises annualized return to 14.49% and Sortino to 1.18, but maximum drawdown is -23.12%. The 10% collar has 14.47% annualized return, 1.17 Sortino, and a -23.31% maximum drawdown. The fixed overlays therefore do not transform SPY. The premium is paid repeatedly, while the payoff is valuable only when the option is sufficiently in the money before expiry. Under daily marks and hedge costs, a small fixed option budget is a path modifier rather than a reliable drawdown solution.

10.2 CPCV and overfit screens

Strategy	Sortino p05	p50	p95	Median max DD	Positive
Equal multi	0.10	1.90	4.50	-15.73%	83.3%
ML downside score, with options	0.96	2.01	4.17	-8.45%	100.0%
Downside score, no options	1.03	1.94	5.75	-10.92%	100.0%
Downside score, with options	1.17	1.66	5.58	-9.83%	100.0%
Options + drawdown control	1.17	1.66	5.58	-9.83%	100.0%
SPY	0.77	1.45	2.38	-16.49%	100.0%

Table 13: CPCV split distribution. Strategies are refit on purged training folds.

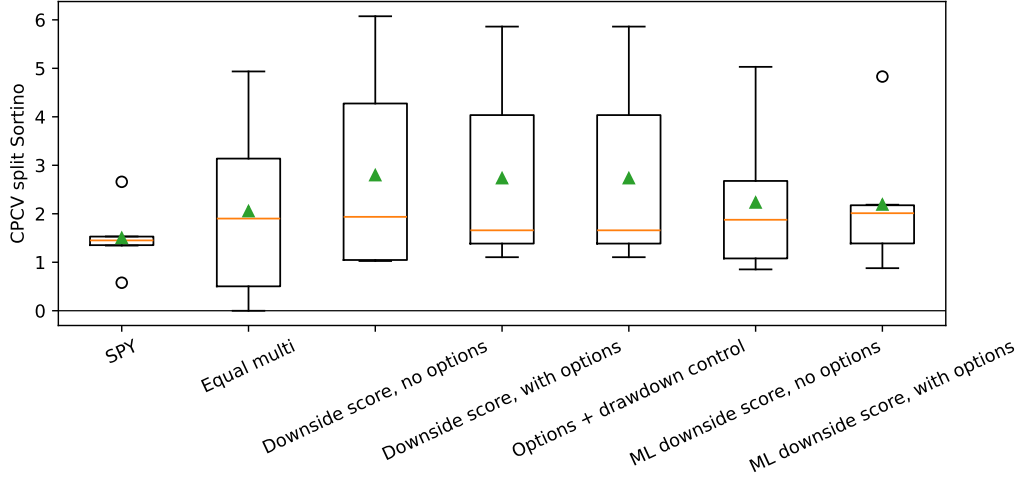


Figure 3: CPCV Sortino distribution across held-out folds.

The CPCV result supports the historical finding without making it final. SPY's median CPCV Sortino is 1.45. Rolling inverse volatility reaches 2.98. The no-option downside-score book reaches 1.94, and the option-enabled book reaches 1.66 with a better median maximum draw-down. CPCV therefore does not bless the option universe unconditionally: it shows that the SPY option overlays can reduce path loss, but the simpler no-option book still has the stronger median Sortino. The displayed non-option and allocator strategies have positive Sortino in every CPCV split, which is partly a feature of the favorable sample and partly a warning that six splits are not a large statistical universe.

Status	PBO	Trials	Metric
ok	0.44	70	sortino

Table 14: Probability of backtest overfitting diagnostic across strategy/config variants.

The PBO diagnostic [8] is 0.44 across 70 trials. That is a warning, not a decorative statistic. It says that once the candidate set includes strategy and configuration variants, the best in-sample Sortino configuration often does not remain the best out of sample. The paper therefore does not treat the PBO screen as confirmation of a single optimal rule. It treats the simpler equal-weight, inverse-volatility, and downside-score results as more credible than a heavily selected configuration only because they are predeclared, low-dimensional, and also survive the realized path, CPCV summaries, cost stress, and simulation screens.

10.3 Simulation path uncertainty

The simulation tables should be read before making any strong option claim. They ask a different question from the realized path: not “what happened from 2018 to 2026,” but how often a strategy with the observed return blocks or volatility clustering would revisit the same drawdown region. This is where the option-enabled book earns part of its keep, and where the no-option book’s higher return must be weighed against path-loss risk.

Strategy	Lag-1	Skew	Ex. kurt.	Recommended simulation
ML downside score, with options	-0.06	0.87	22.77	block bootstrap or egarch
Downside score, no options	-0.02	0.35	9.08	iid empirical resample ok
Downside score, with options	-0.03	0.63	15.06	iid empirical resample ok
Options + drawdown control	-0.02	0.09	8.49	iid empirical resample ok
SPY	-0.13	-0.29	13.07	block bootstrap or egarch

Table 15: Simulation diagnostics. Serial dependence pushes interpretation toward blocks or EWMA paths.

Strategy	Simulation	Sortino p50	Max DD p50	Terminal p05	Terminal p50
ML downside score, with options	block	1.67	-15.29%	1.69	2.71
ML downside score, with options	egarch	1.39	-21.67%	1.62	2.45
Downside score, no options	block	2.14	-14.22%	1.91	3.44
Downside score, no options	egarch	1.71	-20.31%	1.41	3.18
Downside score, with options	block	2.02	-12.71%	1.76	2.88
Downside score, with options	egarch	1.83	-16.66%	1.63	2.77
Options + drawdown control	block	1.82	-14.27%	1.57	2.57
Options + drawdown control	egarch	1.64	-18.39%	1.69	2.52
SPY	block	1.19	-30.60%	1.27	3.19
SPY	egarch	0.80	-43.23%	0.58	2.27

Table 16: Monte Carlo path summaries. EGARCH rows use fitted EGARCH when available and EWMA fallback otherwise.

Strategy	Simulation	DD > 8%	DD > 10%	DD > 15%
ML downside score, with options	block	100.0%	100.0%	53.8%
ML downside score, with options	egarch	100.0%	98.3%	88.3%
Downside score, no options	block	100.0%	92.5%	43.8%
Downside score, no options	egarch	100.0%	98.3%	86.7%
Downside score, with options	block	100.0%	90.0%	23.8%
Downside score, with options	egarch	100.0%	95.0%	61.7%
Options + drawdown control	block	100.0%	93.8%	42.5%
Options + drawdown control	egarch	100.0%	98.3%	68.3%
SPY	block	100.0%	100.0%	98.8%
SPY	egarch	100.0%	100.0%	100.0%

Table 17: Simulated drawdown-breach probabilities.

The simulation layer makes the conclusion more skeptical. Under daily circular block bootstrap, SPY breaches a 10% drawdown in 100.0% of paths. The no-option downside-score book breaches in 92.5% of paths, and the option-enabled book breaches in 90.0%. Under the EWMA/EGARCH-style volatility-clustered simulation, SPY breaches in 100.0% of paths, the no-option book in 98.3%, and the option-enabled book in 95.0%. That does not make the option-enabled book the best portfolio; it says the SPY option sleeve is most visible when the validation method emphasizes clustered downside states. The historical OOS path favored the no-option book on return and Sortino, while the option-enabled book had lower volatility, CVaR, maximum drawdown, and some lower simulated drawdown-breach probabilities. A serious conclusion has to keep both facts.

10.4 ML as forecast estimation

Strategy	Model	Selected	Mean IC	IC p05	IC p95	Folds
ML downside score, no options	ridge alpha 1	no	0.497	0.337	0.625	5
ML downside score, no options	ridge alpha 5	yes	0.499	0.339	0.598	5
ML downside score, no options	lgbm leaves 7	no	0.478	0.355	0.570	5
ML downside score, no options	lgbm leaves 15	no	0.458	0.348	0.552	5
ML downside score, with options	ridge alpha 1	no	0.473	0.415	0.539	5
ML downside score, with options	ridge alpha 5	yes	0.481	0.419	0.562	5
ML downside score, with options	lgbm leaves 7	no	0.433	0.211	0.556	5
ML downside score, with options	lgbm leaves 15	no	0.446	0.220	0.576	5

Table 18: Purged-CV ML forecast validation. Headline admission still depends on portfolio-level validation.

The ML arm does not clear the bar for a headline result. In the realized OOS table, the ML no-option strategy earns 13.55% with a 1.41 Sortino and a -22.72% drawdown. The ML option-enabled strategy earns 12.82% with a 1.53 Sortino and a -20.41% drawdown. The non-option ML book is competitive with simple benchmarks, but both ML books are worse than the simpler non-ML downside-score books on the realized path. The purged-CV forecast table is useful but not decisive: the frozen model choices show some positive IC evidence, yet that evidence does not translate into a better portfolio-level result. That is not a failure to use machine learning. It is the correct use of machine learning: the model remains a documented research arm until forecast value survives portfolio-level validation, not just fold-level IC.

10.5 Regime and option-state evidence

Regime	Strategy	Months	Ann. ret.	Max DD	Hit rate
calm	SPY	41	32.21%	-4.66%	78.0%
calm	Downside score, no options	41	28.79%	-4.17%	80.5%
calm	Downside score, with options	41	25.10%	-3.41%	82.9%
calm	Options + drawdown control	41	24.76%	-3.41%	82.9%
calm	ML downside score, with options	41	20.06%	-3.22%	80.5%
stress	SPY	59	3.19%	-25.58%	59.3%
stress	Downside score, no options	59	6.49%	-21.37%	62.7%
stress	Downside score, with options	59	5.40%	-18.61%	55.9%
stress	Options + drawdown control	59	3.62%	-25.15%	52.5%
stress	ML downside score, with options	59	8.05%	-21.84%	61.0%

Table 19: Regime-conditional returns. Regimes are descriptive cuts, not optimizer inputs.

State	Overlay	Months	Mean	P95 payoff	Positive
calm	SPY 95P	68	-0.23%	-0.08%	0.0%
calm	SPY 95 105 COLLAR	68	-0.36%	0.01%	5.9%
stress	SPY 95P	68	0.09%	2.68%	10.3%
stress	SPY 95 105 COLLAR	68	0.05%	2.71%	10.3%

Table 20: Option overlay payoffs by state bucket.

The regime table explains why the multi-asset result is not just high average return. In stress months, SPY's annualized return is 3.19% with a -25.58% drawdown. The no-option downside-score book earns 6.49% with a -21.37% drawdown, and the option-enabled book earns 5.40% with a -18.61% drawdown. The option-state table gives the important micro result. The put has a negative mean return in calm states and a small positive mean return in stress states; the collar is close to flat in stress states. Both have upper-tail payoffs concentrated in stress. That is why the right question is not whether options make money on average or have upside convexity. The right question is whether their state payoff is worth the repeated premium inside the total portfolio.

10.6 Cost stress

Strategy	Costs	Ann. ret.	Sortino	Max DD
Downside score, no options	1x	15.12%	2.55	-19.87%
Downside score, no options	3x	14.98%	2.52	-19.96%
Downside score, with options	1x	13.07%	2.66	-15.18%
Downside score, with options	3x	12.61%	2.54	-15.36%
Options + drawdown control	1x	11.82%	2.19	-18.04%
Options + drawdown control	3x	11.35%	2.09	-18.58%
ML downside score, with options	1x	12.82%	2.01	-20.41%
ML downside score, with options	3x	11.95%	1.86	-20.88%

Table 21: Cost-multiplier stress on realized path returns.

The cost stress is intentionally simple. Costs are multiplied by one, two, and three using the fixture turnover and asset-level cost assumptions. The non-ML downside-score books remain positive under 3x costs because the monthly turnover and sleeve caps keep the penalty small. This should not be read as proof of capacity. It says the reported fixture result is not erased by a modest cost multiplier. A production book would still need executable futures rolls, OPRA bid-ask fills, VIX-futures liquidity, and capacity curves.

10.7 What the experiment does not prove

The experiment does not prove that the selected downside-score rule will trade well in production. The futures series are continuous proxies. The VIX-futures rule is simplified. The option marks are monthly slice trade closes rather than executable intraday strategy fills. The stock basket is not a point-in-time full S&P 500 portfolio. The OOS sample has 100 monthly returns after the training window, which is enough to challenge the theory but not enough to support broad statistical claims. The correct conclusion is narrower: the framework generates coherent, reproducible, and economically interpretable results on real local data.

11 Robustness and Claim Audit

The main overfit risk is not one parameter. It is the temptation to let a flexible system justify any sleeve after the fact. The paper controls that risk through a claim audit and through validation artifacts. Each claim is tagged as standard source material, adapted model choice, generated empirical result, or original contribution. Exact source claims require page-image verification or a manual recheck against the source PDF. Generated empirical claims require a manifest entry.

Claim	Type	Status
Optimizer decision tree	standard	source anchored
Downside/CVaR/drawdown objectives	standard	source anchored
Cashflow normalization	original framing	proved/sketched here
ML forecast band	adapted	validation caveat
Fixture objective comparison	empirical	generated manifest

Table 22: Claim-audit summary. The full ledger is in docs/claim_ledger.md.

Result 5 (Forecast uncertainty widens admission). *Let an ML model produce a sleeve forecast $\hat{\mu}_i$ with validation error s_i . A robust allocator should admit the sleeve using a haircut such as $\hat{\mu}_i - zs_i$, not the point forecast alone. Better validation narrows the band; leakage, overlap, unstable features, and many-trial search widen it.*

The audit also keeps the writing honest. A research proxy is labeled as a research proxy. A live trading requirement is labeled as a production gap. A theorem sketch is not used to claim live trading alpha. The point is to make the paper harder to overstate.

The paper uses a hierarchy of evidence. The realized OOS path is the first screen because it is the actual historical sequence. CPCV is the second screen because it refits strategies across alternative held-out histories with purging and embargo. Monte Carlo is the third screen because drawdown is path-dependent and needs resampling or volatility-clustered uncertainty checks. ML forecast validation is the fourth screen because forecasts can look useful before leakage, fold dispersion, and repeated-trial selection are charged.

The hierarchy is deliberately asymmetric. A strategy can pass the OOS path and still be rejected if CPCV or simulation reveals fragility. A strategy can look promising in ML validation and still be held out of the headline result if IC intervals cross zero or PBO is poor. This is why the paper presents the ML arm as a research screen rather than a claim of superior allocation.

The same caution applies to source use. Standard definitions and known results are cited to the literature. Exact empirical numbers are cited to generated artifacts. Any source summary that has not been rechecked against the underlying page is allowed to motivate design, but not to support a precise numerical or formula claim. That separation is deliberately conservative: it is better for a paper to admit a narrow, verified claim than to sound complete while resting on unverifiable detail.

12 Production Gap

The clean research object is a domestic cashflow process with a reproducible friction model. The live object is an executable book. That gap is large. A production system would need exchange calendars, broker margin, financing agreements, security master history, corporate actions, borrow and locate rules, option exercise handling, CTD and delivery logic, futures roll execution, FX settlement calendars, curve bootstrapping, quote quality filters, TCA, paper-trading parity, monitoring, and kill switches.

The daily execution layer added here narrows the gap but does not close it. It converts monthly targets into daily close/settlement P&L, records option lifecycle decisions, charges deterministic sleeve costs, uses lagged delta-hedge inputs, and predeclares VIX-futures roll and drawdown-control rules. That is materially more realistic than a monthly return matrix. It is still not a broker simulator: interim option marks can be model-derived, continuous commodity futures remain proxies, and no order book queue or implementation-shortfall model is calibrated from live fills.

The sub-repo is designed so that these gaps can be closed one at a time. The base package defines the instrument, objective, forecast, validation, and reporting interfaces. New data feeds can be added behind the data registry. New solvers can be added behind the optimizer interface. New execution assumptions can be added to the friction profile. The paper does not need to pretend those pieces are already finished.

The investment-committee interpretation is therefore bounded. The framework can say which portfolio construction method is coherent for a given asset mix, which sleeves survive a downside and cost screen, and which production facts remain unknown. It cannot turn a clean research result into live alpha without execution evidence.

There are two different promotions. Promotion to an independent research repository means the paper builds from a clean clone, the fixture reproduction needs no credentials, the generated tables and figures are not hand-edited, the tests cover the cashflow and validation contracts, and optional large data jobs are separated from the base run. Promotion to capital is a different standard. The live system must use point-in-time data snapshots, validated security masters, quote filters, trading calendars, borrow and financing feeds, broker margin, execution assumptions cal-

ibrated from TCA, paper-trading parity, monitoring, and kill switches. A strategy that passes the first promotion but fails the second remains research.

13 Conclusion

The paper began with a practical question: if an investor already owns SPY, when should the book add stocks inside the index, commodities, VIX futures, Treasuries, and SPY options? The answer is not “whenever the in-sample optimizer says so.” The instrument has to be represented as a funded cashflow, marked under the correct convention, forecast under the physical measure, charged for frictions, and tested against downside objectives.

The empirical answer is qualified but useful. Over the February 2018–May 2026 OOS window, the multi-asset portfolios improve the downside path relative to SPY. The no-option downside-score book is the highest-return realized book among the headline allocators, while the option-enabled downside-score book has lower drawdown, volatility, and CVaR but slightly lower realized Sortino. CPCV supports the broad ranking across alternative held-out histories, but the PBO diagnostic is a warning that configuration selection risk remains material. The equal multi-asset portfolio improves Sortino and CVaR but not drawdown, which is an important guardrail. Diversification is doing real work, but asset labels alone are not enough; the paper should not pretend it has identified a unique optimal rule.

The simulation layer makes the conclusion more honest. Block bootstrap strongly favors the downside-score books over SPY on drawdown-breach probabilities. The EWMA volatility-clustered paths still favor the optimized books, but they show that both optimized books can breach 10% drawdown much more often than the realized path suggests. The option-enabled book does not beat the no-option book on realized return or Sortino, but it does improve volatility, CVaR, maximum drawdown, and some simulated drawdown-breach probabilities. That is the economic role of convexity in this sample: it is insurance, not free performance.

The ML arm is not admitted as a headline result. The frozen model choices produce some useful fold-level evidence in the repaired validation run, but the ML allocations still underperform the simpler downside-score books on the realized path. That is the correct standard. Machine learning may help once richer point-in-time features, better option-surface state variables, and longer histories are available, but it cannot bypass purging, embargo, PBO, costs, fold-dispersion checks, and portfolio-level evidence.

Options are the central caveat. The fixed 10% protective-put and collar overlays do not solve SPY’s drawdown problem in the daily-executed sample. They slightly change the path and improve some tail-loss and drawdown moments, but their maximum drawdowns remain close to SPY’s. The option-enabled downside-score book is more useful: it lowers CVaR, volatility, drawdown, and some simulation stress screens, but it gives up annual return. Buying convexity is rational when the premium and hedge cost are worth the improvement in the portfolio’s bad states. It is not rational merely because the payoff is nonlinear.

In practice, the decision rule should be:

admit sleeve $\iff$ robust portfolio contribution after costs is positive.

For equities, that contribution is mostly spot participation and cross-sectional risk. For gold and copper, it is inflation, growth, and commodity-state exposure. For VIX futures, it is stress convexity offset by roll drag. For Treasuries, it is duration and key-rate payoff subject to the inflation regime. For options, it is convex payoff net of premium, skew, spread, theta, and hedge cost. The portfolio objective decides whether those roles are worth owning together.

The limitations are visible. The data are good enough for a research paper but not for a live trading claim. Futures, VIX futures, options, and the constituent basket are all simplified in ways that a production system would have to repair. The OOS sample contains 100 monthly returns after daily execution, enough to challenge the theory but not enough to support broad statistical claims.

The next version should deepen the evidence, not broaden the claims. The natural additions are a full point-in-time S&P constituent reconstruction, executable futures and VIX-futures rolls, OPRA bid/ask execution, observed long-dated single-name option chains, delta-hedged option sleeves, Treasury-futures CTD and key-rate accounting, and richer purged forecast-validation experiments. Those additions should enter through the same objects used here rather than through a new story invented after the fact.

The useful conclusion is therefore practical. Build the cashflow. Charge the friction. Choose the downside objective before fitting the optimizer. Treat options as portfolio-level state-contingent insurance, not as isolated bets. Reject any sleeve whose attractive result depends on free hedging, unpriced roll, stale quotes, or future information. Keep only the portfolio evidence that survives path tests, CPCV, dependence-aware simulations, and cost stress, and state plainly what still has to be true before capital is allocated.

A Proof Sketches

A.1 Cashflow normalization

For each instrument, write all contractual payments, collateral flows, variation margin, premiums, coupons, dividends, funding charges, and hedge trades in domestic currency. Discount each cashflow by the domestic money-market account. The aggregate portfolio cashflow is the sum across instruments because wealth accounting is additive. The only nonlinearity is in the state-dependent payoff or cost map, not in the accounting identity.

A.2 Pricing versus allocation

No-arbitrage pricing changes the measure so normalized prices are martingales under a valid numeraire. Allocation asks for physical-measure expected domestic return after premia, costs, and constraints. The two steps answer different questions. Confusing them either double-counts carry or treats a pricing drift as an investment edge.

A.3 Sortino admission

Take a small sleeve perturbation around the current book. The numerator changes with robust net mean. The denominator changes through the sleeve's covariance with shortfall states $(\tau - R)_+$. The sleeve enters only when the robust mean improvement exceeds the shortfall risk and cost hurdle.

A.4 Drawdown hedge admission

Drawdown depends on the path through the running peak. A hedge that pays in the states setting the maximum drawdown can reduce the path risk even if its unconditional mean is negative. The sign of standalone mean is therefore not sufficient for admission.

A.5 Friction-deflated geometry

Suppose the feasible set $\mathcal{K}$ is convex, the gross domestic payoff map is affine in positions, the cost map C_ϕ is convex state by state, and the payoff ordering allows free disposal. For two feasible

books x_1, x_2 and $\lambda \in [0, 1]$,

$$C_\phi(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda C_\phi(x_1) + (1 - \lambda)C_\phi(x_2).$$

The affine gross payoff and convex cost imply that the free-disposal closure of attainable net payoffs is convex under this ordering. It is generally not a cone because $C_\phi(ax) \neq aC_\phi(x)$ when impact, hedge cost, or capacity penalties are nonlinear. This is the geometric reason that adding a contract to the catalog is not the same as adding a free spanning direction.

A.6 Forecast band

Let the validation error of forecast $\hat{\mu}_i$ be s_i . If the robust decision uses the lower confidence forecast $\hat{\mu}_i - z s_i$, then a larger validation error lowers the admission score one-for-one in z . A model with leakage or unstable labels should therefore face a wider admission band. This is a portfolio consequence of forecast uncertainty, not a claim about one learning algorithm.

B Solver Reductions

This appendix records the optimization problems that the body paper refers to only briefly. The point is not that every desk should run the most complex solver first. The point is that each method should answer a precise investment question and should fail visibly when its inputs are too weak.

B.1 Common scenario notation

Let $x \in \mathbb{R}^n$ be sleeve weights or capital allocations, $r_s \in \mathbb{R}^n$ be the net instrument return vector in scenario s , and p_s be scenario probability. The net portfolio return is

$$R_s(x) = r_s^\top x - C_s(x),$$

where C_s contains spreads, commissions, impact, financing, margin drag, roll cost, hedge cost, and data-quality penalties. A cash sleeve is included explicitly when the portfolio can hold un-invested collateral. A leverage constraint is written as $\|x\|_1 \leq L$, but implementations should replace this with the actual gross, margin, issuer, country, tenor, and liquidity constraints used by the mandate.

The same scenario cube can feed every optimizer. The difference is the objective and the admission rule. This avoids a common research error: comparing optimizers that silently used different costs, calendars, or fill assumptions.

B.2 Mean-variance and robust mean-variance

The classical quadratic problem is

$$\max_x \quad \hat{\mu}^\top x - \frac{\lambda}{2} x^\top \hat{\Sigma} x \quad \text{s.t.} \quad x \in \mathcal{X}.$$

In this paper it is never treated as a default portfolio engine. It is admissible only after shrinkage, explicit constraints, and cost charges are active. If expected return uncertainty is represented by an ellipsoid

$$\mathcal{U}_\mu = \{\mu : \mu = \hat{\mu} + Au, \|u\|_2 \leq \rho\},$$

then the worst-case mean contribution is

$$\inf_{\mu \in \mathcal{U}_\mu} \mu^\top x = \hat{\mu}^\top x - \rho \|A^\top x\|_2.$$

The robust quadratic program becomes

$$\max_x \quad \hat{\mu}^\top x - \rho \|A^\top x\|_2 - \frac{\lambda}{2} x^\top \hat{\Sigma} x - C(x) \quad \text{s.t.} \quad x \in \mathcal{X}.$$

This form makes the economics readable. A sleeve with a high point forecast but poor validation history gets its mean cut before it competes for capital.

B.3 Sortino objective

For target return τ , the sample Sortino objective is

$$S_\tau(x) = \frac{\sum_s p_s R_s(x) - \tau}{(\sum_s p_s (\tau - R_s(x))_+^2)^{1/2}}.$$

The denominator is not volatility. It is the square root of a lower partial moment. Two portfolios with the same variance can therefore rank differently if one loses money in the wrong states. The ratio is finite only when the shortfall moment is positive, or when the implementation uses a documented denominator floor. This is the paper's main organizing criterion.

A practical solver can use a grid or bisection on a candidate ratio η . For fixed η , accept x only if

$$\sum_s p_s R_s(x) - \tau \geq \eta \left(\sum_s p_s z_s^2 \right)^{1/2}, \quad z_s \geq \tau - R_s(x), \quad z_s \geq 0.$$

If $R_s(x)$ is affine, or if the modeled costs preserve a conic representation, the feasibility screen can be written as a second-order-cone program. With more general nonlinear costs it becomes a nonlinear scenario problem. In the small fixture implementation, the same economics are tested by deterministic candidate search so the base reproduction has no heavy solver dependency.

B.4 CVaR loss program

Let $L_s(x) = -R_s(x)$. The Rockafellar–Uryasev representation gives

$$\text{CVaR}_\alpha(L(x)) = \min_{q \in \mathbb{R}} \left[q + \frac{1}{1 - \alpha} \sum_s p_s (L_s(x) - q)_+ \right].$$

The corresponding constrained portfolio problem is

$$\begin{aligned} \max_x \quad & \sum_s p_s R_s(x) \quad \text{s.t.} \quad q + \frac{1}{1 - \alpha} \sum_s p_s u_s \leq \bar{c}, \\ & u_s \geq -R_s(x) - q, \quad u_s \geq 0, \quad x \in \mathcal{X}. \end{aligned}$$

This is a linear or convex program whenever the net return map has the required convexity. It is useful when the mandate cares about expected loss beyond a quantile, not simply the quantile itself.

B.5 Drawdown and CDaR program

Drawdown constraints require a path, not a one-period distribution. In an additive-PnL or linearized wealth approximation, let $W_t(x)$ be cumulative wealth, $P_t(x)$ the running peak, and $D_t(x)$ the drawdown magnitude:

$$P_t(x) \geq P_{t-1}(x), \quad P_t(x) \geq W_t(x), \quad D_t(x) = P_t(x) - W_t(x).$$

For percentage drawdown, replace D_t by $1 - W_t/P_t$, which is nonlinear unless wealth is normalized or linearized. A maximum drawdown cap is $D_t(x) \leq \bar{d}$ for every t . A CDaR cap uses the same auxiliary-variable trick as CVaR:

$$q + \frac{1}{1-\alpha} \sum_t \pi_t v_t \leq \bar{d}, \quad v_t \geq D_t(x) - q, \quad v_t \geq 0.$$

The important desk intuition is that drawdown is path-specific. A hedge can be worth owning even when it loses money on average if it pays during the path segment that sets the portfolio's running peak-to-trough loss.

B.6 Risk parity and inverse volatility

Inverse-volatility sizing uses

$$x_i = \frac{\sigma_i^{-1}}{\sum_j \sigma_j^{-1}}$$

after volatility is estimated on net returns and after instruments that cannot be held at the required scale are excluded. Risk parity equalizes marginal variance contributions:

$$x_i(\Sigma x)_i = x_j(\Sigma x)_j, \quad i, j \in \{1, \dots, n\}.$$

These methods are attractive when expected-return estimates are weak. They still need the same instrument accounting as the advanced methods. A Treasury future, an FX forward, and a listed option cannot be made comparable by volatility alone unless their margin, roll, and hedge conventions are first normalized.

B.7 HRP and clustered allocation

Hierarchical risk parity starts from a distance matrix, clusters similar assets, orders the covariance matrix quasi-diagonally, and recursively allocates risk across clusters. It is a useful baseline because it avoids direct inversion of a noisy covariance matrix. In this paper HRP is treated as an allocator, not as an alpha model. The forecast layer still has to state why any cluster should earn a net premium after costs.

B.8 Black-Litterman-style priors

When desk views exist, they enter as uncertain forecasts rather than hard labels. A compact representation is

$$\mu_{\text{post}} = \mu_0 + \tau \Sigma P^\top (P \tau \Sigma P^\top + \Omega)^{-1} (q - P \mu_0),$$

where μ_0 is the prior, P maps assets to views, q is the view vector, and Ω is view uncertainty. The posterior still flows into the same robust admission screen. A view with a wide Ω should move capital less than a view that survives out-of-sample testing.

B.9 Solver selection rule

The solver menu can be summarized as follows:

Weak alpha	→	cash, inverse vol, risk parity, HRP,
credible alpha	→	robust MVO or Sortino with costs,
tail mandate	→	CVaR, stress floors, drawdown caps,
path mandate	→	CDaR and max-drawdown constraints,
ML forecast	→	forecast only, widened admission bands.

This is the operational meaning of the paper’s claim hierarchy. Complexity is earned by the quality of the data, forecast, and mandate. It is not a substitute for them.

C Instrument Convention Reference

This appendix turns the body paper’s instrument algebra into a desk checklist. The same fields are used by the code in `InstrumentSpec`. Each asset class is specified by payoff, mark, cash account, exposures, frictions, and validation risk.

C.1 Equities and ETFs

An equity position with shares q_t , spot S_t , dividend d_{t+1} , and transaction cost c_t has domestic cash increment

$$\Delta C_{t+1} = q_t(S_{t+1} - S_t) + q_t d_{t+1} - c_t.$$

For foreign equities, spot return and FX return must both be expressed in domestic currency. The exposure map contains equity beta, sector or factor loadings, FX delta when relevant, borrow cost for short positions, and liquidity capacity. The main failure mode is false diversification: many equity sleeves that appear different by ticker can share the same crash beta.

ETFs inherit the same accounting but add creation-redemption, tracking error, fund expense, constituent liquidity, and tax or distribution details. Levered and inverse ETFs are not treated as ordinary equities because daily reset creates path dependence.

C.2 Listed options

A listed option has premium paid today, mark-to-model or exchange mark during the holding period, and terminal payoff

$$(S_T - K)_+ \quad \text{or} \quad (K - S_T)_+$$

for European calls and puts. American exercise, dividends, assignment, and early exercise risk have to be stated separately. The exposure vector includes delta, gamma, vega, theta, rho, jump exposure, implied-volatility surface factor loadings, and liquidity by strike and maturity.

The paper uses no-arbitrage pricing for marks and physical-measure forecasts for allocation. A cheap-looking option is not an allocation edge unless its expected domestic cashflow after hedge costs and spread decay survives validation. Delta hedging is represented by an inner policy with no-trade bands, not by a free continuous replication assumption. The high-level cost proxy is

$$H \approx a \Gamma^2 \sigma^2 \Delta t + b \frac{s}{h} + c h^2,$$

where h is the hedge threshold and s is spread. The formula is deliberately coarse; it reminds the allocator that gamma, realized volatility, and trading cost must be charged before the sleeve competes with simpler instruments.

C.3 Futures

A futures contract is marked to market through variation margin. Its wealth effect is not the same as buying the underlying with cash. For multiplier m and contract price F_t ,

$$\Delta C_{t+1} = q_t m (F_{t+1} - F_t) - \text{commission} - \text{roll cost} - \text{margin drag}.$$

The margin account earns collateral return but also consumes liquidity. Futures exposures include spot beta, curve exposure, roll yield, delivery or expiry calendar, convexity when relevant, and exchange margin. A backtest that ignores roll calendar or margin calls is not testing a futures strategy.

Treasury futures require special care because the contract references a deliverable basket. A simple research proxy may use key-rate duration and a cheapest-to-deliver proxy, but a production implementation must model conversion factors, delivery option, invoice price, repo, and financing.

C.4 Forwards and FX

A forward contract exchanges cashflows at maturity. Under domestic discounting, a foreign-currency payoff is first converted through the FX spot and then discounted by the domestic money-market account. The no-arbitrage forward relation is a pricing constraint, not a statement that FX carry is free alpha.

For FX, the domestic investor holds an exposure to spot, foreign cash return, domestic funding, forward points, and transaction cost:

$$R_{t+1}^{\text{FX}} \approx \Delta s_{t+1} + r_t^f - r_t^d - \text{spread} - \text{roll cost}.$$

The exact implementation should use the contract convention: deliverable forward, NDF, spot-plus-swap roll, or currency future. The exposure map includes FX delta, carry, basis, capital controls when relevant, and funding liquidity.

C.5 Fixed income

A zero-coupon bond paying one unit at T is marked by

$$P(t, T) = \mathbb{E}_t^{\mathbb{Q}} \left[\exp \left(- \int_t^T r_u \, du \right) \right]$$

under the appropriate pricing measure. Allocation then asks whether duration, curve, carry, roll-down, convexity, and liquidity are attractive under the physical measure. The portfolio exposure map should include level, slope, curvature, key-rate DV01, spread duration for credit, inflation beta where relevant, and financing.

A coupon bond is represented as a strip of cashflows. Key-rate exposure is the finite-difference sensitivity of the marked value to curve-node shocks:

$$\text{KRD}_j = - \frac{P(y + \epsilon e_j) - P(y - \epsilon e_j)}{2\epsilon P(y)}.$$

This is more useful for allocation than a single duration when the strategy trades curve shape, butterflies, or inflation/rates regimes.

C.6 Swaps and Treasury future proxies

The first version of this project does not claim to implement a full SOFR multi-curve stack. It uses zero-coupon, key-rate, and Treasury-future proxies. The correct production extension would separate projection curves, discount curves, collateral rate, reset calendars, convexity adjustments, and clearing margin. The paper marks this as a production gap rather than hiding it inside a generic fixed-income bucket.

C.7 Instrument checklist

Instrument	Required convention	Main exposure	Common research error
Equity	Share count, dividend, borrow, FX if foreign	Spot beta, factor beta, FX	Treating correlated equities as diversifiers
ETF	NAV, fee, tracking, distribution, creation basket	Underlying basket, tracking error	Ignoring fund structure and liquidity
Option	Premium, exercise, settlement, vol surface, hedge policy	Delta, gamma, vega, theta	Treating delta hedging as free
Future	Multiplier, margin, roll, expiry, delivery rule	Spot, curve, roll, collateral	Treating notional like cash equity
Forward	Maturity exchange, discounting, collateral, settlement	Carry, basis, FX or commodity spot	Mixing pricing drift with alpha
FX	Spot, forward points, domestic and foreign cash	FX delta, carry, basis	Ignoring funding and roll spread
Bond	Coupon strip, day count, curve, financing	DV01, key-rate DV01, convexity	Using one duration for curve trades
Treasury future	CTD proxy, conversion factor, margin, roll	Duration, basis, delivery option	Ignoring delivery and financing option

Table 23: Instrument convention checks before a sleeve enters the allocator.

C.8 Data-quality score

Every instrument receives a data-quality score before optimization. The score reflects point-in-time availability, stale marks, survivorship, corporate actions, contract roll quality, option quote filters, missing margin history, and whether a source claim has been verified to the required exactness level. A weak score does not always exclude an instrument. It widens forecast uncertainty and raises the admission hurdle.

C.9 Failure-mode log

The research file for each instrument should record the first failure found, not only the final allocation. Examples include: option quote survives mid-price checks but fails open-interest filters; futures roll return disappears after using executable roll windows; an FX carry sleeve works before funding basis and fails after; a bond butterfly is factor neutral but cannot clear financing and transaction-cost stress. This is the style used in the reference papers: the rejection logic is part of the research result.

D Validation and Promotion Reference

This appendix defines the minimum validation contract for promoting the paper into a standalone research repo. The contract is intentionally conservative. A result that passes the fixture run should be treated as a reproducibility screen, not as deployable capital evidence.

D.1 Point-in-time data rule

For every observation used by a forecast or optimizer, the project records:

(observation date, release date, available-to-trade date).

The model may use the value only after the available-to-trade date. Revised economic series, option surfaces, index membership, corporate actions, and contract rolls need explicit availability rules. If the availability rule cannot be reconstructed, the feature is either excluded from the base paper or marked as requiring recheck in the claim ledger.

D.2 Walk-forward split

A default walk-forward experiment has training block T_k , purge block P_k , embargo block E_k , and test block V_k . Training labels whose outcome window overlaps the test block are removed. Observations immediately after the test block are embargoed when leakage can pass through overlapping positions or slow reporting. The fold report includes net return, Sortino, max drawdown, CVaR, turnover, capacity, and realized costs.

D.3 Combinatorial purged cross-validation

The CPCV module partitions the OOS calendar into N chronological groups and tests all $\binom{N}{k}$ combinations of k held-out groups. Training labels whose outcome interval overlaps the test interval are removed, and an embargo removes observations near the test boundary. The resulting split metrics estimate how sensitive the strategy is to the particular OOS path.

The module also assembles CPCV paths. Each group appears in multiple held-out combinations, so the path count is $\binom{N-1}{k-1}$ under the standard CPCV geometry. The fixture configuration uses $N = 4$ and $k = 2$, which is intentionally small for a 100-month OOS sample. Larger promoted studies should increase N once more history is available.

D.4 Probability of backtest overfitting

PBO is computed across strategy/configuration trials by comparing the in-sample winner with its out-of-sample rank across cross-validation partitions. A high value is evidence that selection depends on trying many alternatives and choosing the one that happened to look best in sample. A low value is not proof of edge; it only weakens one overfit channel.

D.5 Reality-preserving Monte Carlo

The simulation module stores three path families. Circular block bootstrap preserves short-run dependence by sampling adjacent blocks. IID empirical resampling is kept as a baseline but is marked weaker when lag-1 autocorrelation or clustering is material. EGARCH is fit when the sample is long enough; otherwise an EWMA residual fallback preserves time-varying volatility without pretending the short fixture sample can support a stable parametric volatility model.

For each strategy and simulation method, the report stores return, Sortino, Sharpe, max drawdown, CVaR, terminal wealth, and breach probabilities for drawdown thresholds. These paths are not alternative histories from a structural model. They are conditional uncertainty screens calibrated to the observed strategy return process.

D.6 Cost multiplier stress

Every result should be recomputed under at least three cost regimes:

$$C^{(1)} = C, \quad C^{(2)} = 2C, \quad C^{(3)} = 3C.$$

The point is not that three times cost is the true worst case. It is a fast screen for fragile research. A sleeve that only works at optimistic cost and fails at modest cost multipliers belongs in the production-gap section, not in the claimed result set.

D.7 Capacity curve

Let A be capital assigned to the strategy. Impact and participation limits create a capacity curve

$$\text{NetEdge}(A) = \text{GrossEdge}(A) - \text{Cost}(A) - \text{Impact}(A).$$

The curve is more informative than a single backtest. It shows whether the idea is a small tactical sleeve, a scalable core allocation, or a research proxy that cannot clear realistic implementation size. The base package includes a simple square-root impact proxy; larger data builds can replace it with venue-specific models.

D.8 Bootstrap uncertainty

For each reported metric m , the project should store a bootstrap interval

$$[q_{0.05}(m^*), q_{0.95}(m^*)]$$

using blocks when serial dependence is material. Point estimates are not enough for portfolio admission. If the interval crosses the rejection threshold, the paper should say that the sleeve is unresolved rather than forcing a positive narrative.

D.9 ML forecast audit

Machine-learning models are restricted to forecasting. They do not bypass the cashflow, cost, hedge, or downside-risk layers. A forecast file must record:

1. feature availability dates and transformations;
2. target construction and overlap horizon;
3. train, purge, embargo, validation, and test windows;
4. hyperparameter search space and trial count;
5. calibration, feature stability, and residual diagnostics;
6. forecast standard error or another uncertainty proxy;
7. admission result after robust haircut and costs.

This converts a model score into an uncertain expected-return input. The optimizer then decides whether that input is strong enough to matter.

D.10 Paper artifact freshness

The paper should never depend on hand-copied tables. A reproduction run writes `artifacts/generated/*.tex`, `artifacts/figures/*.pdf`, and `artifacts/run_manifest.json`. The manifest records package version, config path, fixture hashes when available, generated table paths, figure paths, random seed, and timestamp. The paper QA step checks that generated table files exist and that the prose avoids unsupported claim language.

D.11 Claim ledger statuses

The ledger uses four claim types:

standard	widely established finance or mathematics claim,
adapted	standard result specialized to this cashflow framework,
empirical	result produced by this repo or an attached study,
original	paper-specific modeling contribution or convention.

It also uses exactness statuses:

KB-grounded	supported by the local note route but not page-image checked,
page-verified	checked against the source PDF page image,
repo-generated	produced by the code in this project,
requires recheck	not suitable for final camera-ready numerical claims.

This is deliberately stricter than normal draft writing. It keeps the paper from turning a useful source note into an exact quotation or a trading claim.

D.12 Promotion checklist

Before this sub-repo is promoted, the following checks should pass:

1. `make reproduce` regenerates all paper artifacts without credentials.
2. `make test` passes unit tests for instruments, objectives, hedging, and validation.
3. `make qa` passes the style and claim scans.
4. `make paper` compiles without unresolved references or missing generated files.
5. Every body claim is either cited, ledgered, generated by code, or labeled as a model choice.
6. Every empirical table is generated from code and linked to a manifest.
7. Every asset class has a convention file or fixture row showing cashflow, funding, and costs.
8. Every ML result reports leakage controls and forecast uncertainty.
9. Every hedge result charges spread, threshold, and realized-volatility dependence.
10. Every production gap is visible in the paper rather than hidden in implementation notes.

D.13 Negative evidence

Negative results remain in the research record. A sleeve that fails after costs, a forecast that loses significance after purging, or a hedge that improves drawdown only at impossible turnover still teaches the allocator something. This is one of the main style choices borrowed from the reference papers: a paper can be stronger when it shows where the attractive story breaks.

E Algorithms

E.1 Sleeve admission

1. Normalize each candidate instrument into domestic cashflows and exposures.
2. Estimate expected return, downside covariance, tail loss, drawdown contribution, and capacity.
3. Apply source and model haircuts to forecasts.
4. Charge entry, roll, funding, margin, hedge, and data-quality costs.
5. Admit the sleeve only if robust Sortino or drawdown contribution clears the cost hurdle.

E.2 Base reproduction

1. Load fixture prices and instrument definitions.
2. Build the scenario cube and friction profiles.
3. Generate instrument, exposure, optimizer, and claim-audit tables.
4. Compare inverse-vol, robust mean-variance, Sortino, and CVaR-style allocations.
5. Write artifacts/run_manifest.json.

E.3 Validation protocol

1. Freeze the universe and data-availability rule before fitting.
2. Split the history in chronological order.
3. Purge training observations whose labels overlap the test horizon.
4. Embargo observations immediately after the test block when leakage risk remains.
5. Fit forecasts only on the training block.
6. Build the portfolio with costs and constraints active.
7. Record path returns, costs, drawdowns, tail losses, and capacity.
8. Repeat over folds and report distributions, not only the best path.

E.4 Promotion protocol

1. Keep the fixture reproduction as the always-on smoke test.
2. Add one data feed at a time behind the data registry.
3. Add one instrument convention at a time behind InstrumentSpec.
4. Add tests before using a new convention in the paper.
5. Regenerate artifacts and update the manifest.
6. Update the claim ledger whenever prose relies on a new source or computed result.

F Implementation Interface

The sub-repo exposes stable objects in `src/xasset_portfolio`: `StateDriver`, `InstrumentSpec`, `FrictionProfile`, `ObjectiveSpec`, `ForecastSpec`, `ValidationSpec`, `PortfolioProblem`, `ScenarioCube`, `HedgePolicy`, and `BacktestResult`. The modules are split by responsibility: accounting, instruments, objectives, optimization, hedging, forecasting, validation, costs, data registry, and reporting.

The command-line entrypoint is:

```
python -m xasset_portfolio.run --config configs/base.yml.
```

The Makefile wraps this with `make reproduce`, `make test`, and `make paper`. The base run has no credentialed dependencies and writes only to `artifacts/`.

F.1 Module contract

Module	Responsibility
<code>accounting.py</code>	Domestic cashflow increments and net return charges
<code>instruments.py</code>	Stable dataclasses and asset-class conventions
<code>objectives.py</code>	Sortino, LPM, CVaR, drawdown, and CDaR metrics
<code>optimization.py</code>	Simple and robust allocation rules
<code>hedging.py</code>	Delta hedge notional, no-trade bands, hedge-cost proxy
<code>forecasting.py</code>	Robust forecast haircuts and admission bands
<code>validation.py</code>	Purged splits, bootstrap intervals, capacity curves
<code>data_registry.py</code>	Fixture loading and problem construction
<code>reporting.py</code>	Generated paper tables, figure, and manifest

Table 24: Self-contained package modules.

F.2 Why this is enough for the paper

The package is intentionally small. It does not price every derivative in production detail. It does establish the interfaces the paper claims: instruments are typed, frictions are explicit, objectives are downside-aware, optimizers consume the same scenario cube, hedge costs are not free, validation utilities are point-in-time oriented, and the paper artifacts are regenerated from code. That is the right base for promotion to a larger repo.

G Source Ledger

The full ledger is maintained in `docs/claim_ledger.md`. It records the claim, classification, local knowledge-base route, source exactness status, and paper location. The ledger is not a substitute for the cited books and papers. It is an audit trail that keeps standard source claims separate from the model choices made in this paper.

Area	Route type	Paper use
Optimizer choice	Portfolio management KB	Method gating
Downside risk	Palomar alternative risk measures	Objective hierarchy
Robustness	Palomar robust portfolios	Forecast and covariance haircuts
Options	Seydel BSM and incomplete markets	Greeks and hedge costs
Rates and FX	Shreve numeraires and term structure	Marking conventions
ML validation	Lopez de Prado route	Purging and forecast uncertainty
Backtesting	Portfolio management KB	Costs, capacity, PIT discipline

Table 25: Compact source-ledger map.

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